

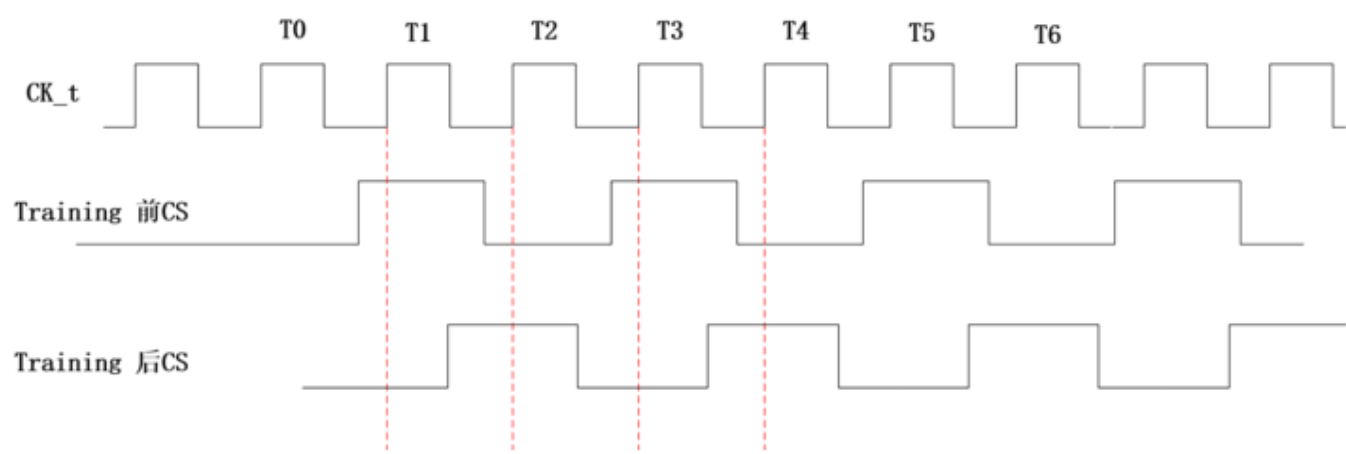
CS/ADR TRAINING原理

1.综述:

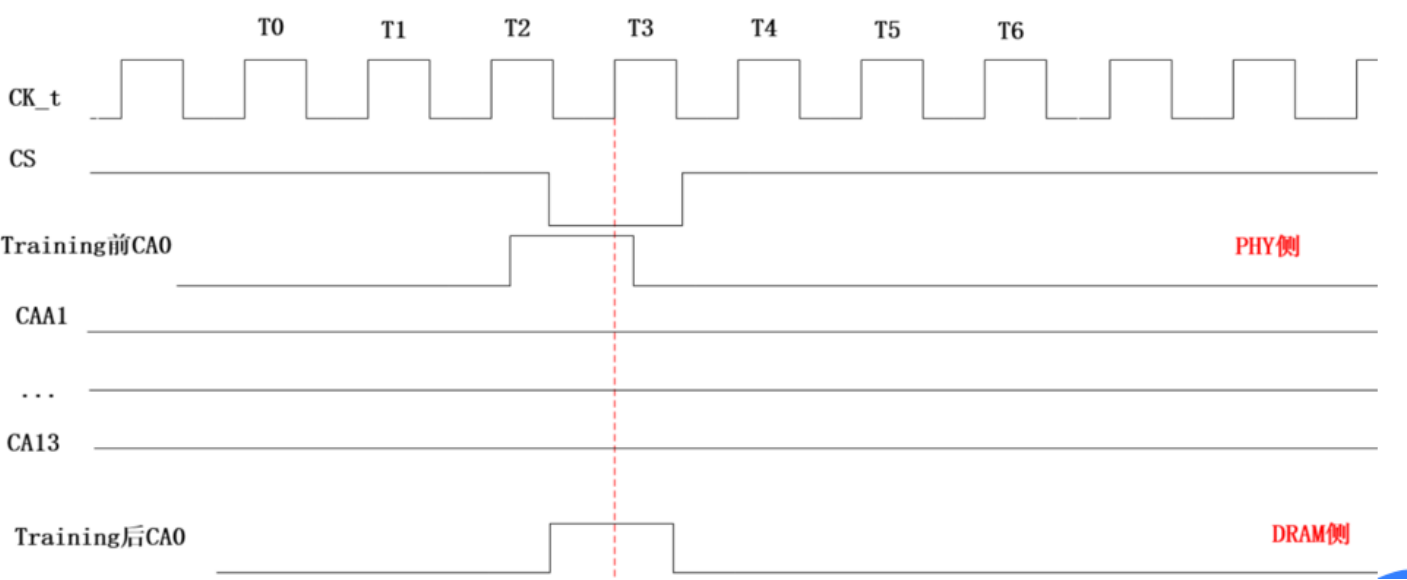
CS/ADR TRAINING在文档中表述为CSLVL TRAINING,该TRAINING项旨在调整CLK/CS/ADDRESS三者之间的相位关系。

其最终的相位调整结果为:

1.1 CS相位:



1.2 ADR相位:



2.CS TRAINING流程:

2.1 找到ADDRESS 3相对于CS初始值0x255的最优值

2.1.1 初始化CS/ADR延迟:

其中：

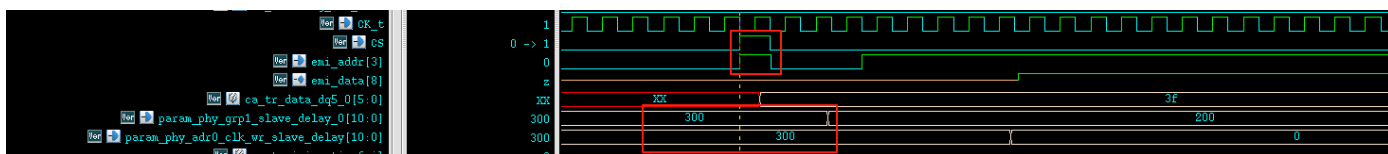
CK_t为采样时钟,CS为片选信号,emi_addr[3]为ADDRESS A3,emi_data[9]为DQ9,DQ9记录CLK对CS+ADR的采样结果,当CLK采样到CS/ADR均为1时返回1,CLK采样到CS为1,ADR为0时返回0,其余情况返回X态

param_phy_grp1_slave_delay_0记录CS0的延迟值

param_phy_adr0_clk_wr_slave_delay记录ADDRESS3的延迟值

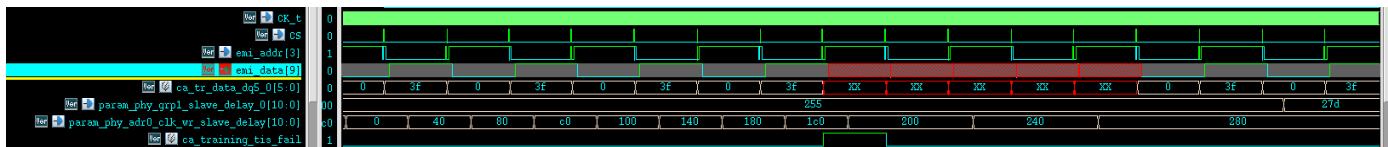
TRAINING开始时,给CS/ADR赋予相同的初始值,使得其在最开始的时候相位对齐,此时DO9返回1

因此PCB上,最好保证ADDRESS与CS线相位对齐

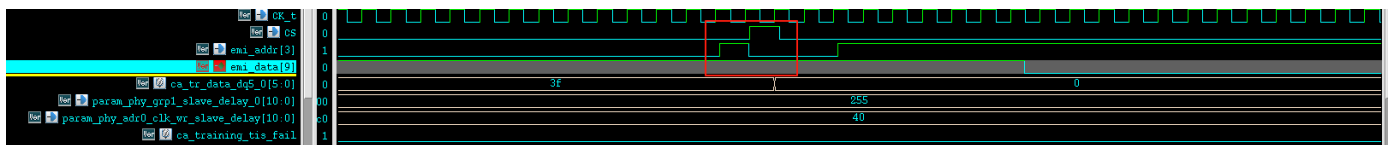


2.1.2 重新设置CS初始值+ADR延迟值清0

此时CS延迟为0x255,ADR为0x0,此时CS出现在ADR的最右边,CLK采值为0,则DQ9返回0



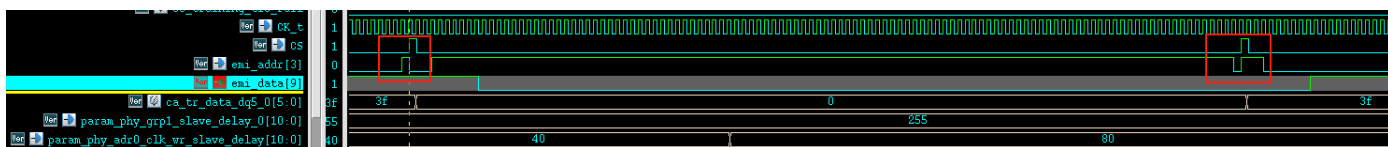
CS出现在ADR的最右边



放大CS对ADDRESS的采样场景:

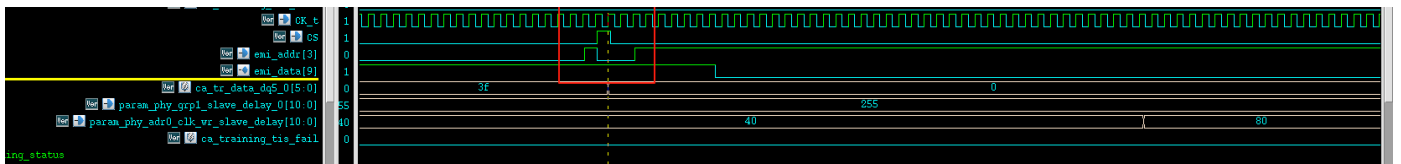
可以看到,每增大一次ADDRESS延迟,发生一次CLK对CS/ADR的采样,

每一次采样,发生一次对ADR为1的采样(左边红框),ADR为0的采样(右边红框)



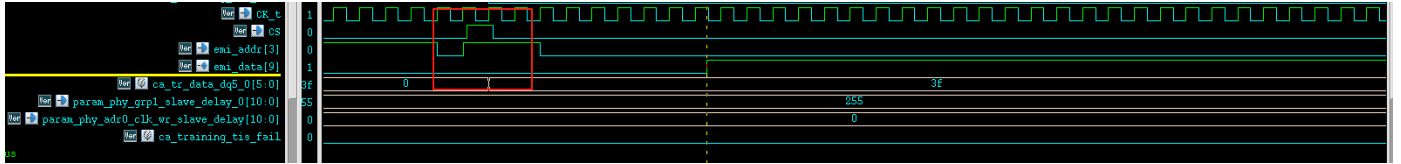
左边红框处采样:

采样到0,返回0

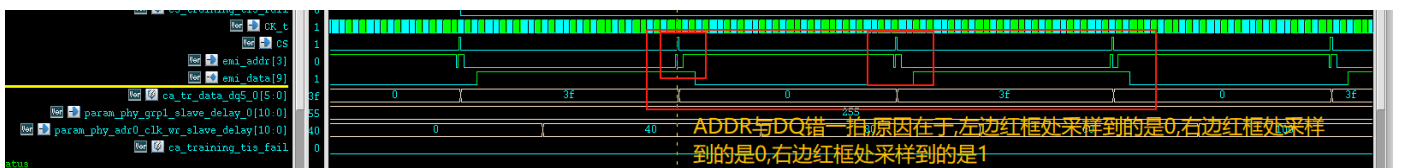


右边红框处采样:

采样到1,返回1

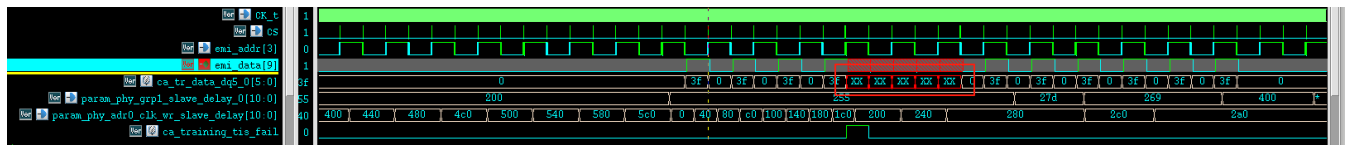


而实际上我们希望,左边采样到1,右边采样到0,因为需要继续增大ADDRESS的延迟,直到能够采样到左1右0



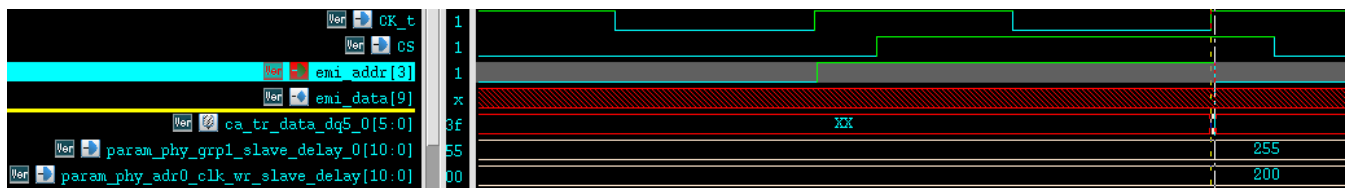
2.1.3 持续增大ADR的延迟,直到CLK采样到ADR为X态

持续增大ADR的延迟,此时DQ9返回X态,返回X态的原因是CLK与ADDRESS的保持时间不足

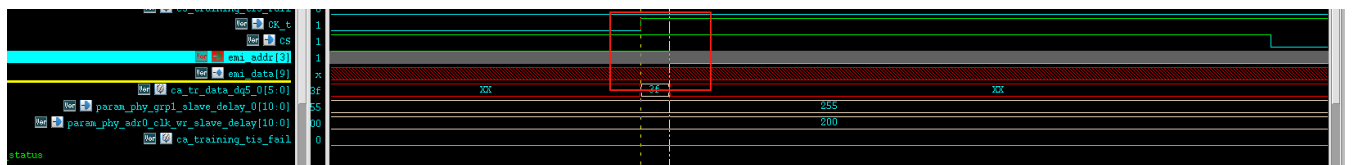


返回X态的采样:

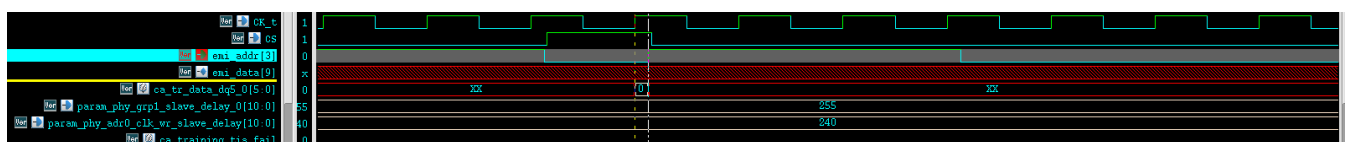
左边



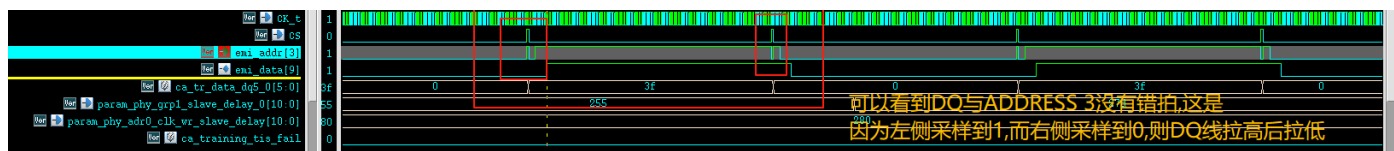
放大:



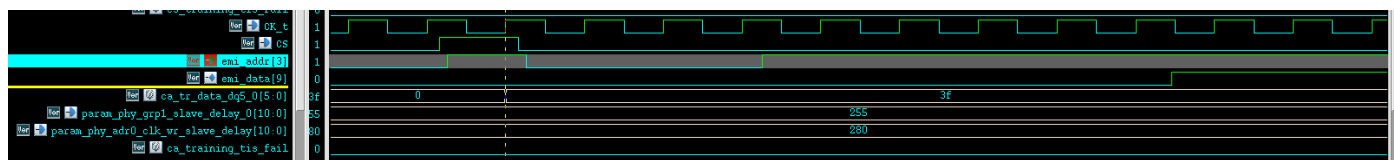
右边



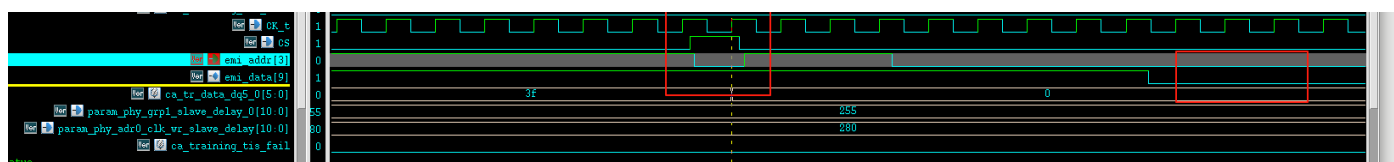
2.1.4 持续增大ADR的延迟,直到DQ9返回1



左侧:



右侧:



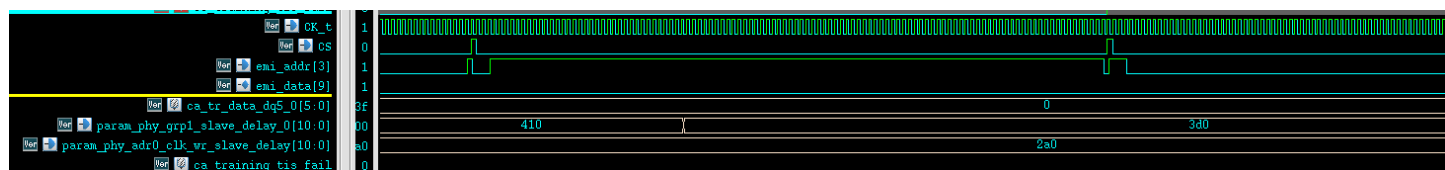
当DQ9第一次采集到1后,再增加若干拍ADR的延迟,则我们认为找到了ADDRESS 3相对于CS初始值0x255的最优值,此时开始搜寻CS的最小保持时间,即左边界

2.2 找到CS的建立时间

2.2.1 保持ADR不动,将CS的初始延迟值翻倍

保持ADR不动,将CS的初始延迟值翻倍,此时相当于再次将CS移动到ADR的最右边:

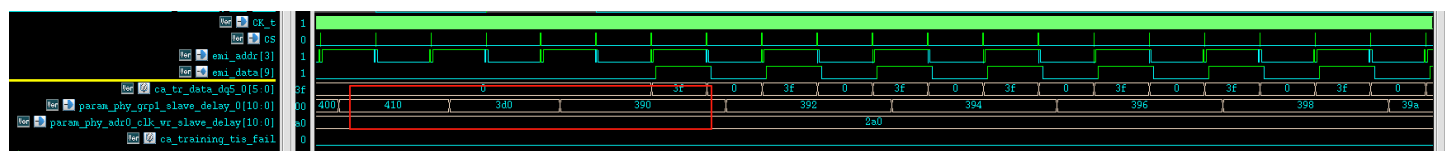
此时CS采样ADR为0:



初始值从0x255调整至0x410,此时CS再次来到ADDRESS的最右边

2.2.2 持续减少CS延迟,直到CS满足相对于CLK的建立时间:

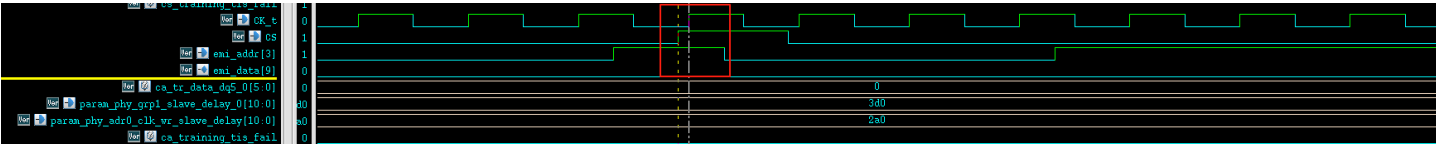
减小CS的延迟,直到满足建立时间:



满足建立时间前:

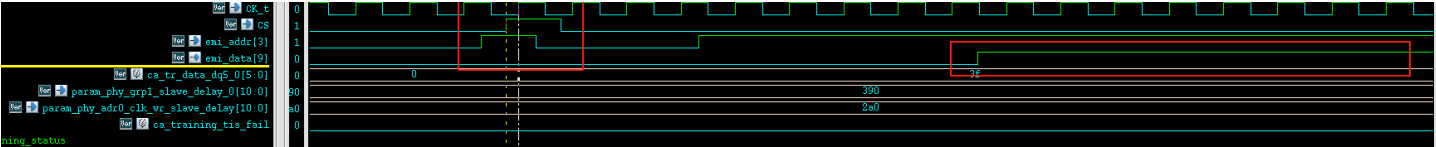
左侧:

此时DQ返回0



减小CS的延迟到满足建立时间后;

此时DQ返回1:



2.3增加CS的建立时间,直到找到极限建立时间,即左边界